

2. The roots of the equation

$$2x^3 - 3x^2 + 12x + 7 = 0$$

are  $\alpha$ ,  $\beta$  and  $\gamma$

Without solving the equation,

(a) write down the value of each of

$$\alpha + \beta + \gamma \quad \alpha\beta + \alpha\gamma + \beta\gamma \quad \alpha\beta\gamma \quad (1)$$

(b) Use the answers to part (a) to determine the value of

(i)  $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma}$

(ii)  $(\alpha - 1)(\beta - 1)(\gamma - 1)$

(iii)  $\alpha^2 + \beta^2 + \gamma^2$  (7)



4. Prove that, for  $n \in \mathbb{Z}$ ,  $n \geq 0$

$$\sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)} = \frac{(n+a)(n+b)}{c(n+2)(n+3)}$$

where  $a$ ,  $b$  and  $c$  are integers to be found.

(5)

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3. The cubic equation

$$ax^3 + bx^2 - 19x - b = 0$$

where  $a$  and  $b$  are constants, has roots  $\alpha$ ,  $\beta$  and  $\gamma$

The cubic equation

$$w^3 - 9w^2 - 97w + c = 0$$

where  $c$  is a constant, has roots  $(4\alpha - 1)$ ,  $(4\beta - 1)$  and  $(4\gamma - 1)$

Without solving either cubic equation, determine the value of  $a$ , the value of  $b$  and the value of  $c$ .

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1. The cubic equation

$$x^3 - 7x^2 - 12x + 6 = 0$$

has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

Without solving the equation, determine a cubic equation whose roots are  $(\alpha + 2)$ ,  $(\beta + 2)$  and  $(\gamma + 2)$ , giving your answer in the form  $w^3 + pw^2 + qw + r = 0$ , where  $p$ ,  $q$  and  $r$  are integers to be found.

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7.

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

(a) Explain why, for  $n \in \mathbb{N}$

$$\sum_{r=1}^{2n} (-1)^r f(r) = \sum_{r=1}^n (f(2r) - f(2r-1))$$

for any function  $f(r)$ .

(2)

(b) Use the standard summation formulae to show that, for  $n \in \mathbb{N}$

$$\sum_{r=1}^{2n} r((-1)^r + 2r)^2 = n(2n+1)(8n^2 + 4n + 5)$$

(6)

(c) Hence evaluate

$$\sum_{r=14}^{50} r((-1)^r + 2r)^2$$

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2. The roots of the equation

$$x^3 - 2x^2 + 4x - 5 = 0$$

are  $p$ ,  $q$  and  $r$ .

Without solving the equation, find the value of

(i)  $\frac{2}{p} + \frac{2}{q} + \frac{2}{r}$

(ii)  $(p - 4)(q - 4)(r - 4)$

(iii)  $p^3 + q^3 + r^3$

(8)

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4. (a) Use de Moivre's theorem to prove that

$$\sin 7\theta = 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta \quad (5)$$

- (b) Hence find the distinct roots of the equation

$$1 + 7x - 56x^3 + 112x^5 - 64x^7 = 0 \quad (5)$$

giving your answer to 3 decimal places where appropriate.

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1. Given that

$$z_1 = 3 \left( \cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right)$$

$$z_2 = \sqrt{2} \left( \cos\left(\frac{\pi}{12}\right) - i \sin\left(\frac{\pi}{12}\right) \right)$$

(a) write down the exact value of

(i)  $|z_1 z_2|$

(ii)  $\arg(z_1 z_2)$

(2)

Given that  $w = z_1 z_2$  and that  $\arg(w^n) = 0$ , where  $n \in \mathbb{Z}^+$

(b) determine

(i) the smallest positive value of  $n$

(ii) the corresponding value of  $|w^n|$

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3.

$$f(x) = \arcsin x \quad -1 \leq x \leq 1$$

- (a) Determine the first two non-zero terms, in ascending powers of  $x$ , of the Maclaurin series for  $f(x)$ , giving each coefficient in its simplest form.

(4)

- (b) Substitute  $x = \frac{1}{2}$  into the answer to part (a) and hence find an approximate value for  $\pi$

Give your answer in the form  $\frac{p}{q}$  where  $p$  and  $q$  are integers to be determined.

(2)

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4. In this question you may assume the results for

$$\sum_{r=1}^n r^3, \sum_{r=1}^n r^2 \text{ and } \sum_{r=1}^n r$$

(a) Show that the sum of the cubes of the first  $n$  positive odd numbers is

$$n^2(2n^2 - 1) \quad (5)$$

The sum of the cubes of 10 consecutive positive odd numbers is 99800

(b) Use the answer to part (a) to determine the smallest of these 10 consecutive positive odd numbers.

(4)

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9. (a) Given that  $|z| < 1$ , write down the sum of the infinite series

$$1 + z + z^2 + z^3 + \dots \quad (1)$$

- (b) Given that  $z = \frac{1}{2}(\cos \theta + i \sin \theta)$ ,

- (i) use the answer to part (a), and de Moivre's theorem or otherwise, to prove that

$$\frac{1}{2} \sin \theta + \frac{1}{4} \sin 2\theta + \frac{1}{8} \sin 3\theta + \dots = \frac{2 \sin \theta}{5 - 4 \cos \theta} \quad (5)$$

- (ii) show that the sum of the infinite series  $1 + z + z^2 + z^3 + \dots$  cannot be purely imaginary, giving a reason for your answer.

(2)

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6. The cubic equation

$$4x^3 + px^2 - 14x + q = 0$$

where  $p$  and  $q$  are real positive constants, has roots  $\alpha$ ,  $\beta$  and  $\gamma$

Given that  $\alpha^2 + \beta^2 + \gamma^2 = 16$

(a) show that  $p = 12$

(3)

Given that  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{14}{3}$

(b) determine the value of  $q$

(3)

Without solving the cubic equation,

(c) determine the value of  $(\alpha - 1)(\beta - 1)(\gamma - 1)$

(4)

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2. (a) Write down the Maclaurin series of  $e^x$ , in ascending power of  $x$ , up to and including the term in  $x^3$

(1)

- (b) Hence, without differentiating, determine the Maclaurin series of

$$e^{(e^x - 1)}$$

in ascending powers of  $x$ , up to and including the term in  $x^3$ , giving each coefficient in simplest form.

(5)

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